## 12.5 Integral Calculus

### 12.5.1 Integration

Differential calculus measures the rate of change of functions. In Economics it is also necessary to reverse the process of differentiation and find the function F(x) whose rate of change has been given. This is called integration. The function F(x) is termed an integral or anti- derivative of the function f(x).

The integral of a function f(x) is expressed mathematically as

$$\int f(x)dx = F(x)+C$$

Here the left hand side of the equation is read "the integral of f(x) with respect to x" The symbol ∫ is an integral sign, f(x) is integrand, C is the constant of integration, and F(x)+c is an indefinite integral. It is so called because, as a function of x, which is here unspecified, it can assume many values.

### 12.5.2 Meaning

If the differential coefficient of $F(x)$ with respect to x is f(x), then an integral of f(x) with respect to x is F(x). It is a reverse process of differentiation. In symbols:

$$\text{If } \frac{d[F(x)]}{d_x} = f(x), \text{ then } \int f(x)dx = F(x)+C$$

Following points need to be remembered:

**a.** ∫ is used to denote the process of integration. In fact, this symbol is an elongated 'S' denoting sum.

**b.** The differential symbol 'dx' is written by the side of the function to be integrated.

**c.** $\int f(x)dx = F(x)+C$, C is the integral constant

$\int f(x)dx$ means, integration of f(x) with respect to x.

### 12.5.3 Basic Rule of Integration

**(i)** Power Rule $\int x^n dx = \frac{x^{(n+1)}}{n+1} + C$

**(ii)** $\int k.dx = kx + c$, where k is a constant

**(iii)** $\int a.x^n dx = a\int x^n dx$

**Example 12.22**

$$\int 4x^3 dx = 4\int x^3 dx$$
$$= 4\frac{x^{3+1}}{3+1} + c$$
$$= 4\frac{x^4}{4} + c$$
$$= x^4 + c$$

**Example12.23**

$$\int (x^2 + x - 1)dx = \int x^2 dx + \int x dx - \int dx$$
$$= \frac{x^{2+1}}{2+1} + \frac{x^{1+1}}{1+1} - x + c$$
$$= \frac{x^3}{3} + \frac{x^2}{2} - x + c$$

**Example 12.24**

$$\int 5dx = 5x + c$$

**Example12.25**

$$\int 4xdx = 4\frac{x^{1+1}}{1+1} + c$$
$$= 4\frac{x^2}{2} + c$$
$$= 2x^2 + c$$



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